Analytics & Big Data

Session 2: Exploratory data analysis

Prof. Dr. Gerit Wagner

(2026-03-23)

🎯 Learning Objectives

- **Describe** key characteristics of variables and patterns between variables.
- **Explore** data sets by using appropriate descriptive and visual techniques in Python (*exercise*).
- **Explain** the role of exploratory data analysis in business decision-making.

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Foundations

> People are not very good at looking at a column of numbers or a whole data table and then determining important characteristics of the data. EDA techniques have been devised as an aid in this situation.

The role of EDA

Exploratory Data Analysis (EDA) is the process of **examining and summarizing data to understand its main characteristics before building analytical models.**

EDA typically involves:

- **Descriptive statistics** (e.g., mean, variance, frequencies)
- **Data quality checks** (e.g., missing values, outliers)
- **Visualization techniques** (e.g., histograms, boxplots, scatterplots)

Goal: gain intuition about the data, detect patterns or anomalies, and prepare the data for further analysis or modeling.

The following forms of EDA are typically distinguished:

	Univariate	**Multivariate**
Non-graphical	mean, median frequencies	cross-tabs correlations
Graphical	histograms dot plots	scatter plots grouped dot plots

Data types

Conducting EDA and representing the “real world” in computer requires us to use appropriate data types. Typically, we distinguish four fundamental data types, known as measurement scales:

Qualitative description of objects

- Nominal (special case: binary)
- Ordinal

Quantitative description objects

- Interval
- Ratio

Qualitative: Nominal scale

- Values of a nominal (aka. categorical) attribute are symbols or names of things
- Each value represents some kind of category, code, or state.
- This scale assumes existence of a finite number of equivalency classes, where each class is named or labeled.
- The values do not have any meaningful order.

Examples:

- Hair color = {black, blond, brown, grey, red, white, …}
- Postal codes = {96047, 96048, …}

Special case of the nominal scale: Binary scale

- Nominal attribute with only 2 states (0/1, or TRUE / FALSE) – also called Boolean
- 0/FALSE typically means absence and 1/TRUE presence

It is recommended to assign 1 to the most important, and often rare, outcome that is subject to investigation (e.g. people that purchase a product, have a disease)

Examples:

- Smoking: 1 indicates patients in a trial that smoke, 0 otherwise.
- Purchase: 1 indicates that a person purchased a product, 0 otherwise.

Qualitative: Ordinal scale[1]

- A categorical attribute with values that have a meaningful order, but the magnitude between successive values is not known
- The ordinal scale is useful for registering subjective assessments and things that cannot be measured objectively (often used in surveys for ratings)

Values cannot be multiplied or added, even if the numbers belong to the same scale.

Relations between values:

- Transitivity: If A>B, B>C, then A>C,
- Symmetry: : If A>B, then B<A

Examples:

- School grades = {A < B < C < D < E < F} or {1 < 2 < 3 < 4 < 5 < 6}
- Places in a competition = {1st, 2nd, 3rd, 4th, …}

1. Also known as a “rank scale”.

Interval scale (quantitative)

- A numeric measureable attribute in the form of integer or real values. The distance between the numbers or units on this scale is equal over all levels of the scale. Values of the interval scale have no natural “zero” point.
- Invariant under a linear transformation $ax + b$, $a > 0$, $b \geq 0$
- Although the sum of two interval-scale measurements is not meaningful by itself, their average can be computed

Examples:

- Celsius scale of temperature (there is no natural zero, 0°C is arbitrarily defined), a Celsius value c can be linearly transformed to a Fahrenheit value f: $f = 9/5 * c + 32$
- Dates and time (e.g., conversion from Julian to Gregorian calendar is possible)

Ratio scale (quantitative)

- Numeric values with a meaningful zero point (e.g., “zero” means absence)
- All arithmetic rules and functions can be applied (addition, subtraction)

Examples:

- Money, person’s age, market share, quantities purchased, speed, …
- Kelvin (K) temperature (has a true zero-point (0 °K=− 273.15 ℃): It is the point at which the particles that comprise matter have zero kinetic energy

Possible mathematical operations for each scale type

Due to the different mathematical properties, not all statistical measures can be computed on different measurement scales:

Scale	Mathematical operations possible	Mode	Frequency	Percentiles / Median	Mean / Variance / SD
Nominal	=, ≠	X	X		
Ordinal	=, ≠, <, >	X	X	X	
Interval	=, ≠, <, >, linear transformation	X	(X)	X	X
Ratio	=, ≠, <, >, *, /, +, −	X	(X)	X	X

x. The frequency (relative / absolute) may be calculated for a numeric variable, but makes only sense when the number of possible values is low

Data preparation: Formatting

Often, the way data is **stored** determines which mathematical operations are possible.

Example: timestamps stored as strings

user_id	timestamp
101	“2025-03-12 14:23:05”
102	“2025-03-12 18:02:11”
103	“2025-03-13 09:15:42”

If treated as a **string**, we can only perform operations such as:

- equality checks ($=$ / $\neq$)
- simple filtering or sorting

This **severely limits analysis**.

Format conversions help transform raw data into representations suitable for EDA and analytical models:

Representation	Example	Possible analysis
Numeric timestamp	1741789385	time differences, time series
Hour of day	14	activity patterns during the day
Day of week	Wednesday	weekday vs. weekend behavior
Day of year	71	seasonal patterns

Note

Data preparation transforms raw data into **formats that allow meaningful mathematical operations and analysis**.

Data quality checks

Before conducting analysis, we must ensure that the **data is fit for use** (Strong et al., 1997).

Common data quality issues include:

- **Missing values** (e.g., incomplete historical market data)
- **Outdated values** (e.g., customer address has changed)
- **Outliers or implausible values** (e.g., age = 250)
- **Inconsistent formats or units** (e.g., dates stored as text, mixed currencies)
- **Duplicate records** (e.g., customers with multiple accounts)
- **Measurement or data entry errors** (e.g., human data entry, broken sensors)

Typical data preparation actions

- **Correct** values (e.g., fix formatting or units)
- **Remove** problematic observations
- **Impute** missing values
- **Collect additional data** if necessary

> In many real-world analytics projects, **data preparation and cleaning can take up to ~80% of the total effort.**

Univariate exploratory data analysis

Descriptive statistics

How to describe data (not the data type)?

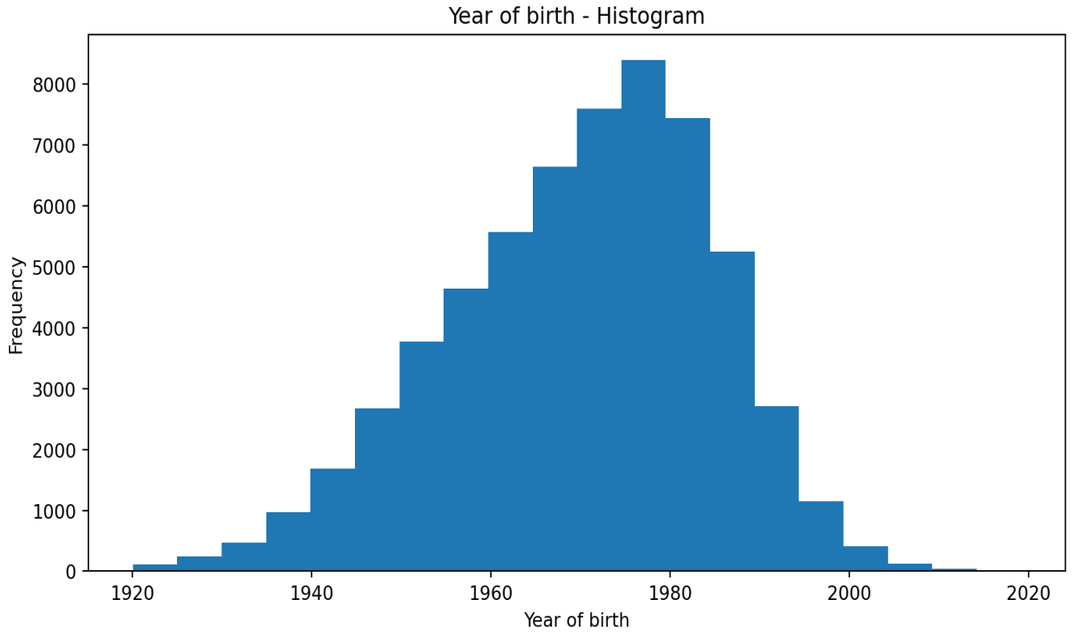
Customer ID	Name	Year of birth	Tariff
100216	Kevin Meyer	1983	A
271692	Lars Knopp	1963	B
892615	Anton Albert	1954	C
331625	Peter Pan	1988	D
…	…	…	…

What are the "key figures" of a distribution?

- Graphics work well for single variables, but for comparison of different variables and their distributions and later working with the data, numeric indicators are necessary
- Descriptive statistics provide us with numbers describing the characteristics of a distribution
- For qualitative / categorical data
  - **Mode**
  - **Relative frequency**
- For quantitative / numerical data
  - **Mean** (also known as **Expected Value**) – describing the position of the center
  - **Variance** (or **Standard Deviation**) – describing the spread
  - **Percentiles** (also known as quantiles / quintiles) – more detailed figures on the distribution

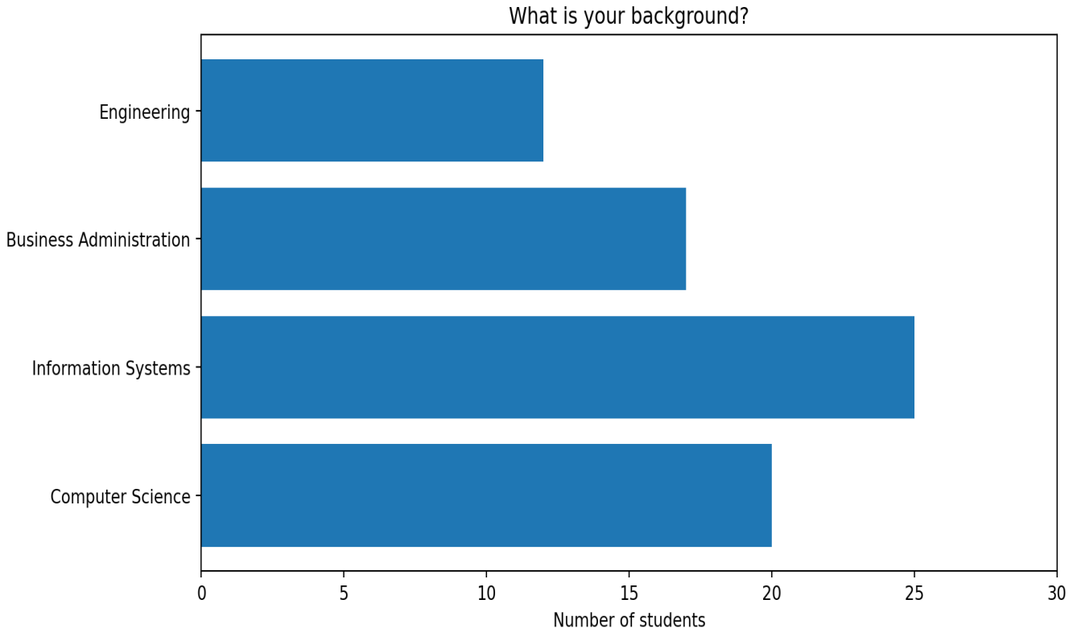
Describing data using descriptive statistics

- Basic statistical descriptions can be used to identify characteristics of the data and highlight which data values should be treated as noise or outliers.
- A good way to get an impression of continuous / numeric data is the **histogram**.
- The graph shows the range of observations on the horizontal axis, with a bar showing how many times each value occurred in the data set.



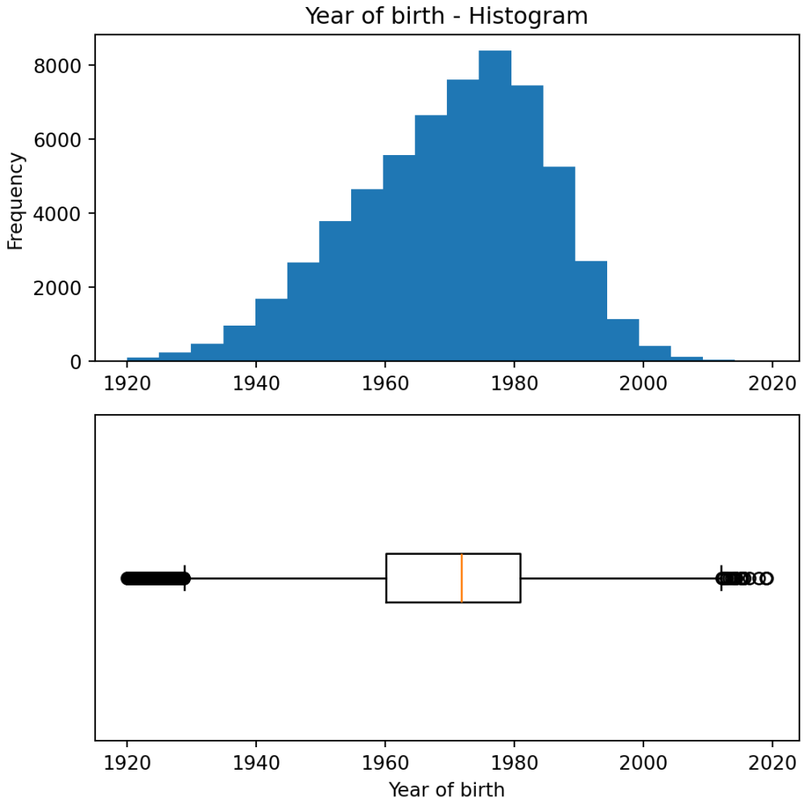
Mode

- The mode is the value that appears most in a set of data values
- The highest mode also has the highest relative frequency (see next slide)
- Example: On a party, you meet many other students, and you ask what they are studying.



Boxplot

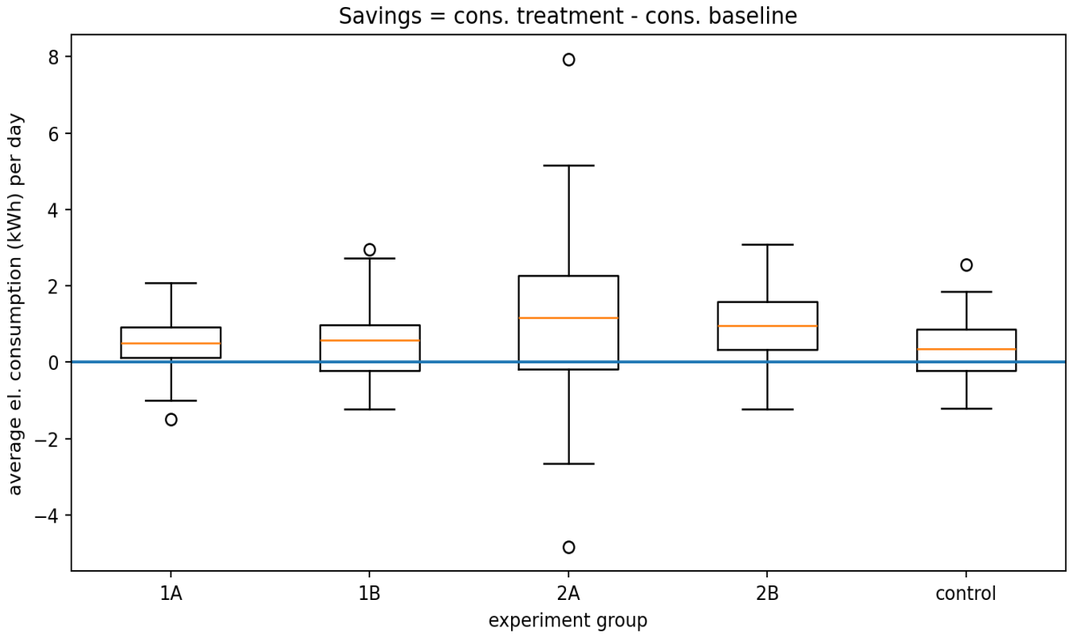
- The boxplot is a condensed illustration of a distribution
- It consists of
  - Median (thick line)
  - 25% / 75% percentiles
  - “Whiskers”: the quartiles ± 1.5 × IQR
  - Outliers that are higher/lower than the whiskers



Boxplots can be used to compare different distributions

Example:

- In an experiment, households received different types of feedback on their electricity consumption
- The “control” group received no feedback
- The plot shows boxplots of the savings in electricity consumption after three weeks with feedback
- One can – for example – see that there is a higher spread in group 2A compared to the others



Multivariate exploratory data analysis

Multivariate Non-Graphical EDA

Multivariate non-graphical EDA techniques generally show the relationship between two or more variables in the form of either cross-tabulation for categorical variables or correlation statistics for numerical variables.

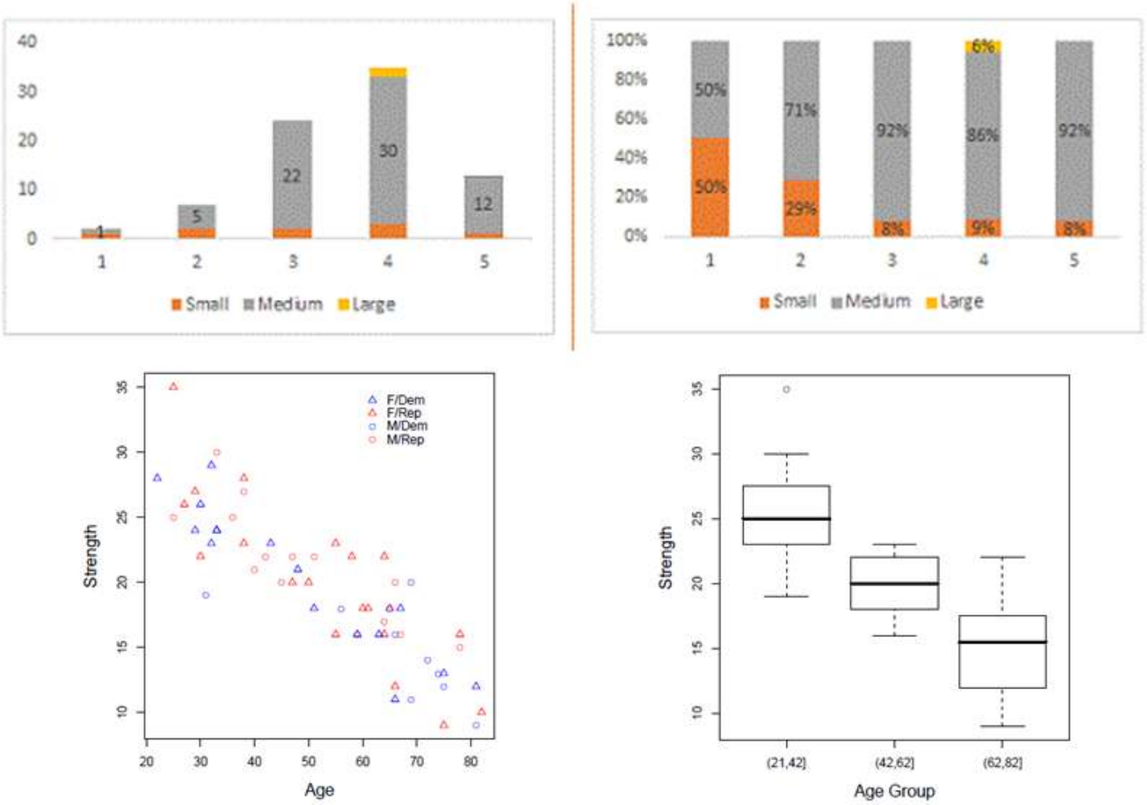
Frequency Row Pct	Product Category 1	2	3	4	5	Total
Small	1 11.11	2 22.22	2 22.22	3 33.33	1 11.11	9
Medium	1 1.43	5 7.14	22 31.43	30 42.86	12 17.14	70
Large	0 0.00	0 0.00	0 0.00	2 100.00	0 0.00	2
Total	2	7	24	35	13	81

Frequency Missing = 77

	V2	V3	V4	V5	V6
V2	-0.388				
V3	0.549	0.189			
V4	0.824	0.758	-0.619		
V5	0.802	0.833	-0.469	-0.946	
V6	-0.186	-0.659	0.349	0.555	0.649

Multivariate Graphical EDA

Multivariate graphical EDA techniques are scatterplots for numerical variables, Barcharts for categorical variables, or Boxplots for mixed types.



Bivariate statistics: Correlation

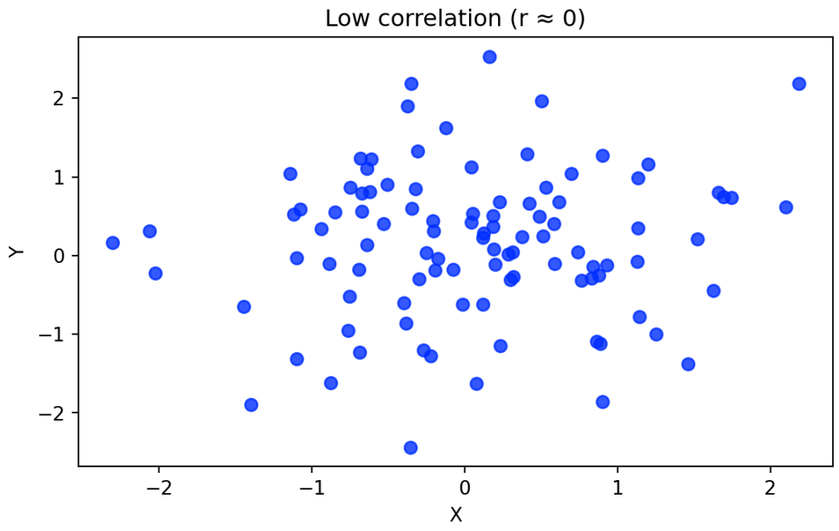
Correlation measures how strongly two variables move together. Suppose we observe two variables for the same observations:

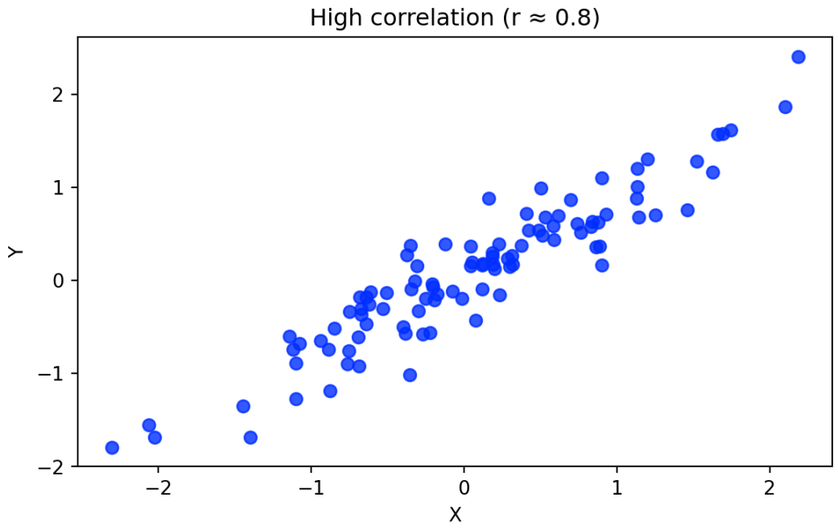
$$X = (x_1, x_2, \ldots, x_n)$$

$$Y = (y_1, y_2, \ldots, y_n)$$

The **Pearson correlation coefficient** is calculated as:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

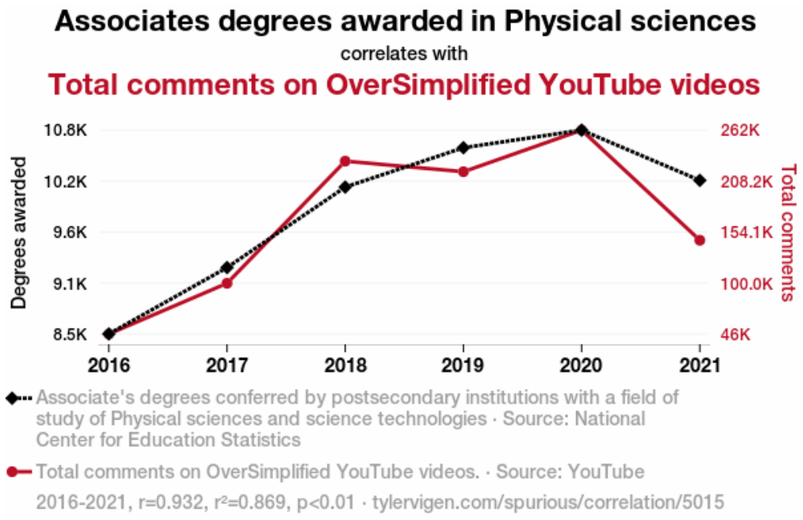


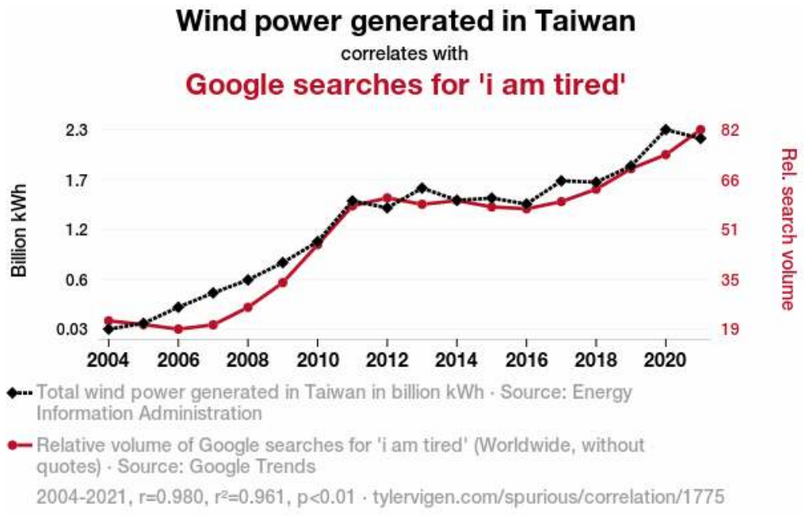


Correlation vs. Causality

Correlation: Two data series behave “similar”

Causality: Principle of cause and effect





Correlation does not imply causation!

> But: Sometimes it is better to predict something even if we cannot explain it instead of doing nothing!

These and more online: https://www.tylervigen.com/spurious-correlations

Clustering in EDA

In **multivariate exploratory data analysis**, we often want to understand whether **observations form natural groups**.

Clustering is a technique that automatically groups **similar observations** based on their characteristics.

Idea: Objects within the same cluster should be **similar to each other**, but **different from objects in other clusters**.

Examples

- **Customer segmentation** in marketing
- **Grouping similar products** based on purchasing patterns
- **Identifying behavioral patterns** in users
- **Detecting anomalies** in transactions

Introductory Example

Assume you are a wholesale distributor and each row of your dataset corresponds to a customer showing the following attributes:

1. FRESH: annual spending on fresh products (Continuous);
2. MILK: annual spending on milk products (Continuous);
3. GROCERY: annual spending on grocery products (Continuous);
4. FROZEN: annual spending on frozen products (Continuous);
5. DETERGENTS_PAPER: annual spending on detergents and paper products (Continuous);
6. DELICATESSEN: annual spending on delicatessen products (Continuous);
7. CHANNEL: customers buying channel (Nominal);
8. REGION: customers region (Nominal)

> Your goal is to segment the users. That means finding similar types of users and bunching them together.
> Why would you want to do this?
> You might want to give different users different experiences. Marketing often does this; for example, to offer toner to people who are known to own printers.
> You might have a model that works better for specific groups. Or you might have different models for different groups.

Clustering: Overview

Clustering groups **observations into clusters based on similarity** across **multiple variables**.

Input

- Each observation is described by **multiple attributes (multivariate data)**
- Example: customer spending on different product categories

$$x_i = (x_{i1}, x_{i2}, \ldots, x_{im})$$

Similarity between observations is typically measured using a **distance metric** (e.g., Euclidean distance).

Types of clustering algorithms

- **Partitioning methods** (e.g., K-Means)
- **Hierarchical methods** (agglomerative, divisive)
- **Density-based methods** (e.g., DBSCAN)

⚠ **Important**

Clustering algorithms will **always produce clusters**, even if the data has **no real structure**.

Therefore, clusters must be **evaluated and interpreted carefully**.

Partitioning Cluster Methods

The partitioning cluster methods divide the data into a predetermined number of clusters. The most popular technique is the K-Means algorithm.

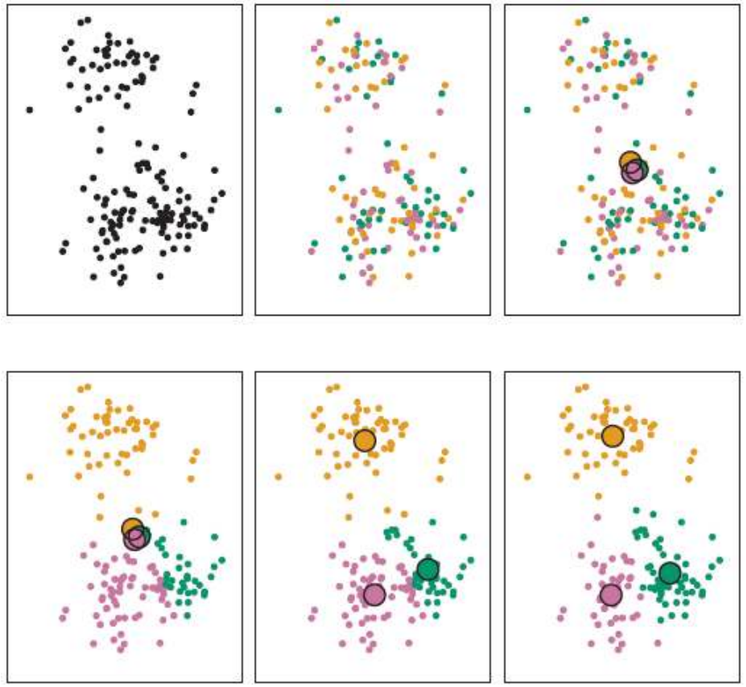
Given a set of observations (x1, x2,…, xn), where each observation is a m-dimensional real vector, k-means clustering aims to partition the n observations into k ≤ n segments S = {S1, S2,…, Sk} so as to minimize the within-cluster sum of squares (WCSS).

The objective is to find where is the mean of points in Si.

> **Procedure of K-Means:**

> Step 1: Randomly partition the data objects into k clusters.
> Step 2: Calculate the cluster centroids.
> Step 3: Calculate the distance from every data point to all centroids.
> Step 4: If a data point is closest to its own centroid, leave it where it is. If the data point is not closest to its own centroid, assign it to the cluster with the closest centroid.
> Step 5: Repeat the step 2 to 4 until a complete pass through of all the data points results in no data point changing from one cluster to another.

Example of a K-Means Cluster Analysis



Between cluster variance:

Within cluster variance:

Finding the Optimal Number of Clusters (I)

The aim of the cluster analysis is the segmentation of objects into clusters, which are preferably homogeneous in it selves and heterogeneous to each other. The less variance exists within the clusters and the more variance exists between the clusters, the better is the number of clusters.

Total variance:

Accumulated variance within the k clusters:

This results in the variance between the clusters:

with n = number of objects

m = number of attributes

Finding the Optimal Number of Clusters (II)

If you put V in on the ordinate and the number of cluster k on the abscissa, it often results in a curve with one or several kinks. At the point where exists the (first) significant kink, you can find the optimal number of clusters:

Total variance V_tot

Between cluster variance V_betw

Within cluster variance V_in

Number of clusters

Finding the Optimal Number of Clusters (III)

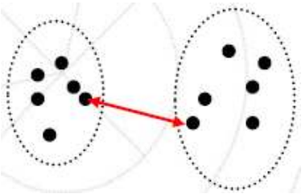
> Instead of visually identifying the optimal cluster number, we can calculate the distances from the points on the elbow curve to a straight line linking the first and the last point on the curve. The cluster number with the largest distance is then chosen as the one with the strongest kink.

Hierarchical Cluster Methods

- There are two types of hierarchical cluster methods:
  - Agglomerative hierarchical clustering is a bottom-up clustering method. It starts with every single data object in a single cluster. Then, in each iteration, it agglomerates (merges) the closest pair of clusters by satisfying some similarity criteria, until all of the data is in one cluster.
  - Divisive hierarchical clustering is a top-down clustering method. It works in a similar way to agglomerative clustering but in the opposite direction. This method starts with a single cluster containing all data objects, and then successively splits resulting clusters until only clusters of individual data objects remain.

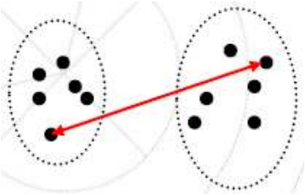
Process of the Hierarchical Cluster Analysis

Measuring Similarity between Clusters (I)

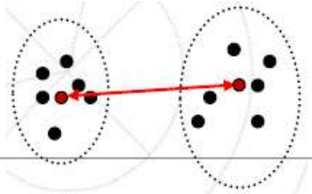


Distance between two clusters is the distance between the closest points:

Complete Linkage:



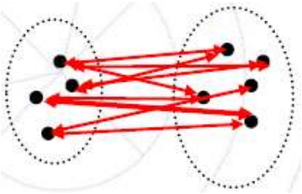
Distance between two clusters is the distance between the farthest pair of points:



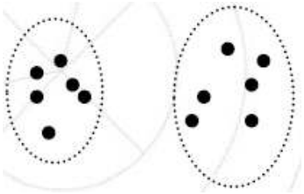
Measuring Similarity between Clusters (II)

Average Linkage:

Distance between clusters is the average distance between the cluster points:



Ward's Method / Minimum Variance Method (only Agglomerative):



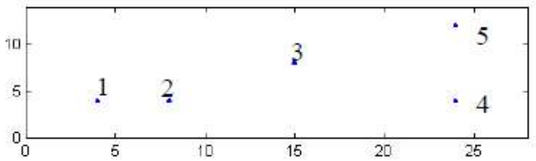
Ward's minimum variance criterion minimizes the total within-cluster variance. At each step the pair of clusters is merged that leads to minimum increase in total within-cluster variance after merging. This can be calculated as the square of the distance between cluster means divided by the sum of the reciprocals of the number of observations in each cluster:

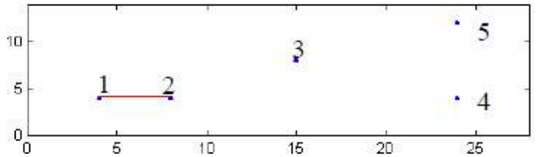
Single Linkage Example (I)

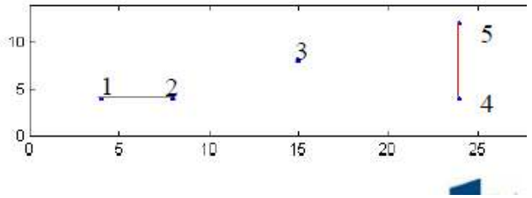
	x	y
1	4	4
2	8	4
3	15	8
4	24	4

	1	2	3	4	5
1	-	4	11.7	20	21.5
2	4	-	8.1	16	17.9
3	11.7	8.1	-	9.8	9.8

	1,2	3	4	5
1,2	-	8.1	16	17.9
3	8.1	-	9.8	9.8
4	16	9.8	-	8
5	17.9	9.8	8	-



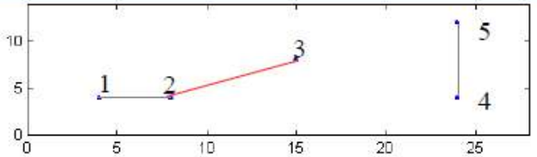




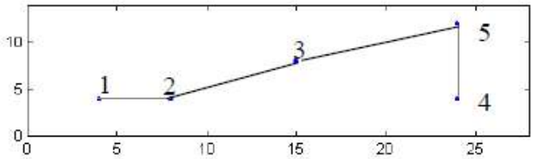
— Source: Fred, Ana: Unsupervised Learning, Universidade Técnica de Lisboa

Single Linkage Example (II)

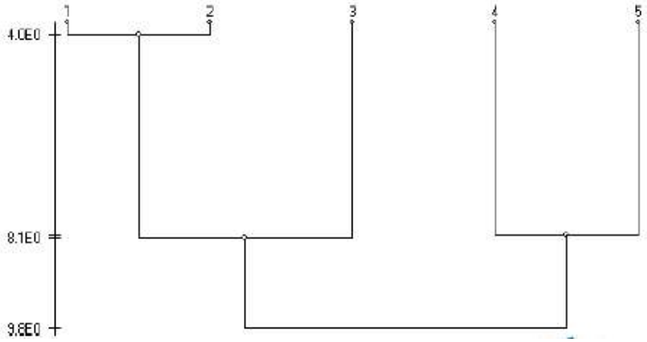
	1,2	3	4,5
1,2	-	8.1	16
3	8.1	-	9.8
4,5	16	9.8	-



	1,2,3	4,5
1,2,3	-	9.8
4,5	9.8	-



Dendrogram



— Source: Fred, Ana: Unsupervised Learning, Universidade Técnica de Lisboa

Dendrogram

A dendrogram is a tree diagram frequently used to illustrate the arrangement of the clusters produced by hierarchical clustering. The y-axis represents the value of this distance metric (e.g. euclidean distance) between the clusters.

In a dendrogram the widths of the horizontal lines give an impression about the dissimilarity of the merging object. Thus, a good cluster number might be at a point from where the width of the following horizontal lines is significantly smaller in length. The red line in the graph below shows such a point:

Counting the points that cut this line might be a good answer for the number of clusters the data can have. It is the number 6 in this case.

Summary

TODO

References

Strong, D. M., Lee, Y. W., & Wang, R. Y. (1997). Data quality in context. *Communications of the ACM*, *40*(5), 103–110. https://doi.org/10.1145/253769.253804